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Project Report: CertiChain

A Decentralized Blockchain-Based Academic Credential Verification System

1. Introduction

In an era where digital transformation is accelerating, the integrity of academic and professional credentials faces significant challenges, including fraud and unauthorized document alteration. This project, **CertiChain**, introduces a robust, decentralized platform designed to bridge the gap between traditional credential issuance and modern blockchain technology. By leveraging Web 3.0, CertiChain ensures that academic certificates are immutable, transparent, and instantly verifiable.

2. Project Objectives

Prevent Fraud: Eliminate certificate forgery by using cryptographic hashes stored on the blockchain.

Enhance Transparency: Provide an open, verifiable audit trail for credential issuance.

Efficiency: Reduce the time and cost associated with manual verification processes for employers and institutions.

Accessibility: Create a user-friendly interface that simplifies the interaction between issuing authorities and verification entities.

3. Technical Stack

The platform is built using a modern 3-tier architecture:

Blockchain Layer: Solidity (Smart Contracts) deployed via Hardhat on an Ethereum-based network.

Backend Layer: Node.js and Express.js, handling API requests and blockchain interactions via ethers.js.

Frontend Layer: HTML, Tailwind CSS, and vanilla JavaScript to provide a responsive and intuitive user experience.

4. System Methodology

The workflow follows a decentralized logic:

1.Issuance: The issuing authority creates a certificate record, and its cryptographic hash is generated and permanently stored on the blockchain.

2.Storage: Digital copies are managed through the backend, linked to the unique blockchain-stored hash.

3.Verification: A requesting entity uploads the certificate file; the system re-calculates the hash and compares it against the blockchain record. If they match, the certificate is deemed authentic.

5. Challenges and Solutions

During the development of CertiChain, several technical hurdles were overcome:

API Routing: Successfully implemented a RESTful API structure to allow seamless communication between the frontend and the blockchain backend.

CORS Configuration: Addressed cross-origin communication issues by configuring strict middleware to allow secure frontend-to-server interaction.

Blockchain Integration: Synchronized the ethers.js provider with the local hardhat environment to ensure real-time transaction monitoring and data retrieval.

6. Conclusion and Future Work

CertiChain provides a scalable and secure solution to the problem of credential verification. By integrating blockchain, we have ensured a "single source of truth" for academic records.

Future enhancements will include:

Scalability: Migrating from a local network to a public testnet (e.g., Sepolia).

Privacy: Implementing Zero-Knowledge Proofs (ZKP) to allow verification without revealing personal data.

Payment Integration: Developing a built-in crypto-payment gateway for service fees.